GammaScalp research memo

Question. The Black-Scholes argument says an option can be replicated by trading the stock continuously, for free. Nobody can do either. How much risk is left when hedging is discrete, costly, and done with the wrong volatility, and do the classical correction formulas predict it?

Method. One short (or long) three-month at-the-money call on a $100 stock, 20% volatility, 3% rate, premium $4.3576. Every study states a prediction from the literature, simulates seeded paths, and reports both numbers. 12,640,000 hedging runs in total (a run is one hedger on one price path; some studies put two hedgers or several rules on the same paths). All figures below are synthetic: they test the formulas and the code, not any market.

1. Discrete hedging: risk falls like 1/sqrt(N)

Derman and Kamal (1999) predict a hedging error with mean zero and standard deviation `sqrt(pi/4) * vega * sigma / sqrt(N)` for N evenly spaced hedges.

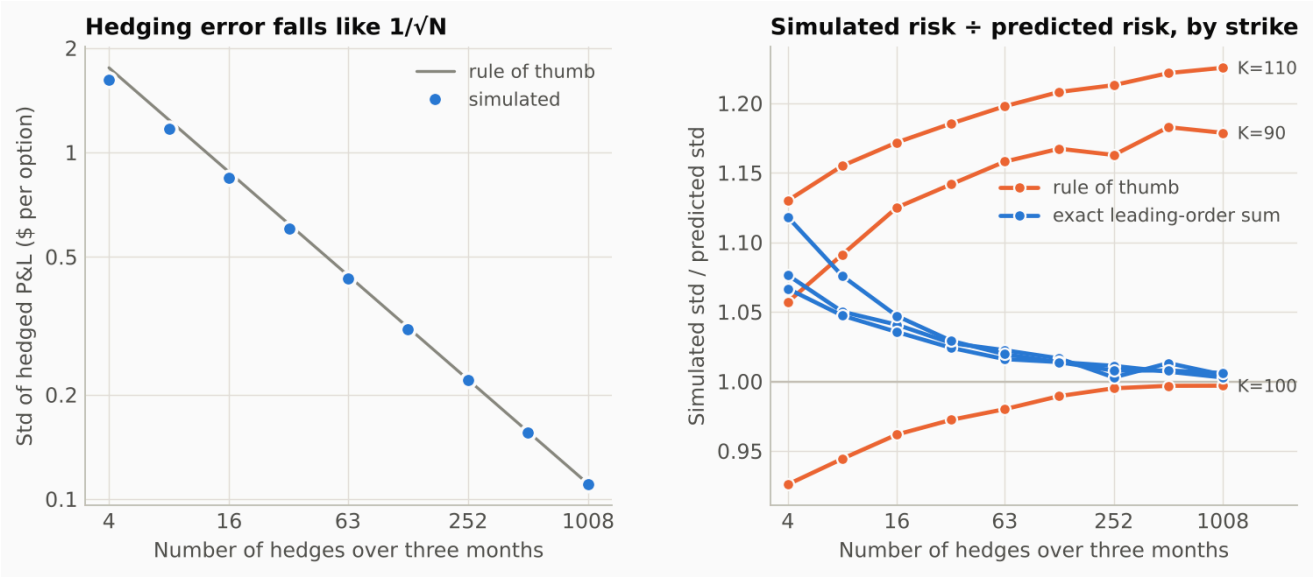
Hedges	Mean P&L	± s.e.	Std	Predicted	Ratio	Std / premium
4	-0.0131	0.0036	1.6240	1.7540	0.926	37.3%
8	-0.0048	0.0026	1.1712	1.2403	0.944	26.9%
16	-0.0001	0.0019	0.8435	0.8770	0.962	19.4%
32	+0.0004	0.0013	0.6030	0.6201	0.972	13.8%
63	-0.0006	0.0010	0.4331	0.4420	0.980	9.9%
126	-0.0013	0.0007	0.3092	0.3125	0.989	7.1%
252	+0.0002	0.0005	0.2199	0.2210	0.995	5.0%
504	-0.0001	0.0003	0.1558	0.1563	0.997	3.6%
1008	-0.0001	0.0002	0.1101	0.1105	0.997	2.5%

- A log-log regression of std on N (16 hedges and up) gives a slope of **-0.491 ± 0.001**; theory says -0.5.
- At 63 hedges (daily) the simulated std is 98.0% of the formula; at 1,008 hedges it is 99.7%. The formula is an asymptotic one and overstates risk when hedges are few.
- Daily hedging still leaves a standard deviation of about 10% of the premium.
- The mean is within two standard errors of zero except at 4 hedges (-0.30% of premium). The sign is the one the next subsection predicts for so few hedges.

The rule of thumb is an at-the-money result. The same experiment at strikes 90 and 110 shows it understating risk: at 1,008 hedges the simulated standard deviation is 18% to 23% above the formula, and the gap does not close with more hedges. The step it skips can be done exactly: each interval contributes `1/2 * sigma^4 * dt^2 * E[(gamma * S^2)^2]` of variance, and summing that over the hedge dates by quadrature (`leading_order_std`) gives a prediction for any strike.

Strike	Hedges	Simulated std	Rule of thumb	Ratio	Exact sum	Ratio
90	63	0.2576	0.2224	1.158	0.2519	1.022
90	1008	0.0655	0.0556	1.179	0.0652	1.005
100	63	0.4331	0.4420	0.980	0.4263	1.016
100	1008	0.1101	0.1105	0.997	0.1098	1.003
110	63	0.3787	0.3161	1.198	0.3714	1.020
110	1008	0.0969	0.0790	1.226	0.0963	1.006

At 1,008 hedges the exact sum is within 0.6% of the simulation at all three strikes; at 63 hedges, within 2.2%. It is a leading-order result, so it too drifts when hedges are few.



A second-order effect: the drift does matter a little

In continuous time a delta hedger is indifferent to the stock's drift. With N discrete hedges the squared move over one interval averages `sigma^2 dt + (mu - r)^2 dt^2`, so a short-gamma position pays a small extra amount. Integrating gives a prediction for a short option: `mean = -(mu - r)^2 * (T / N) * E[integral of 1/2 gamma S^2]`.

Drift	Hedges	Mean P&L	± s.e.	Predicted	% of premium
-0.50	63	-0.0443	0.0008	-0.0434	-1.02%
-0.50	252	-0.0104	0.0004	-0.0109	-0.24%
+0.00	63	+0.0004	0.0010	-0.0002	+0.01%
+0.00	252	-0.0011	0.0005	-0.0000	-0.02%
+0.50	63	-0.0351	0.0008	-0.0350	-0.81%
+0.50	252	-0.0087	0.0004	-0.0087	-0.20%
+1.00	63	-0.0936	0.0006	-0.0929	-2.15%
+1.00	252	-0.0235	0.0003	-0.0232	-0.54%

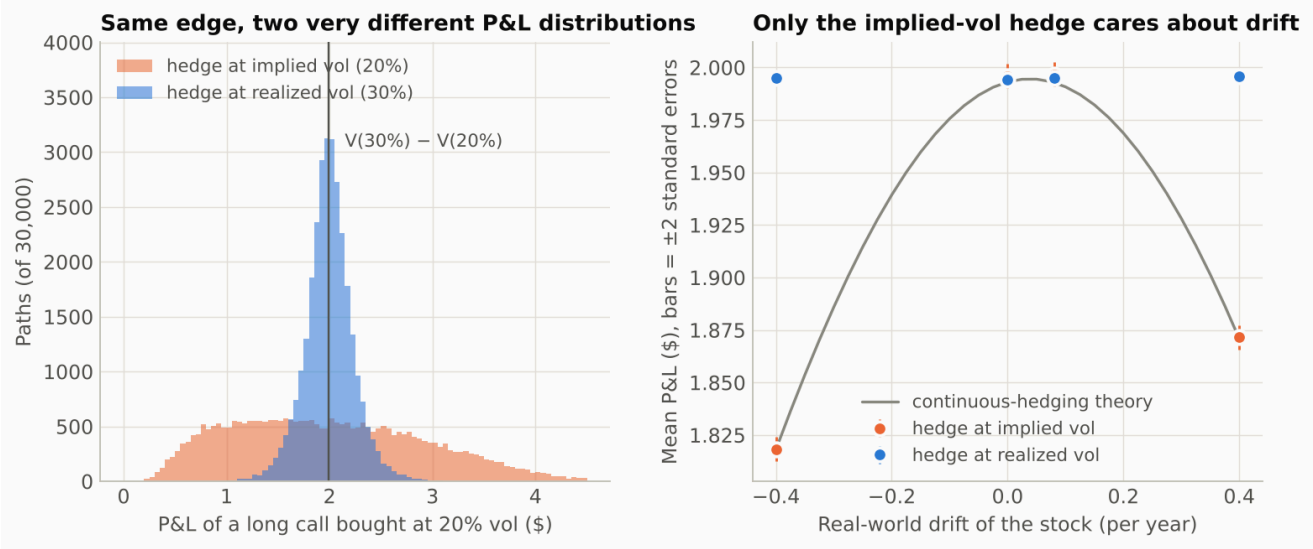
The largest gap between simulation and this prediction is 2.1 standard errors. For realistic drifts the effect is invisible; it becomes about 2.1% of premium only at a drift of 100% a year with daily hedging.

2. Buying volatility cheap: where the money comes from

A call is bought at 20% implied volatility; the stock then realizes 30%. The value of that edge is `V(30%) - V(20%)` = $1.9945 at expiry. Two hedgers trade the same paths (Ahmad and Wilmott, 2005).

Drift	Hedge at realized: mean	std	Hedge at implied: mean	± s.e.	theory	std	5% to 95%
-0.40	1.9950	0.155	1.8183	0.0029	1.8181	0.917	0.56 to 3.47
+0.00	1.9943	0.166	1.9958	0.0029	1.9931	0.922	0.64 to 3.59
+0.08	1.9948	0.167	1.9967	0.0029	1.9928	0.921	0.64 to 3.58
+0.40	1.9958	0.158	1.8715	0.0029	1.8709	0.927	0.57 to 3.52

- Hedging at **realized** volatility locks in the edge: the mean equals `V(30%) - V(20%)` at every drift, and the spread around it is only the discrete-hedging noise of study 1.
- Hedging at **implied** volatility earns the gamma rent `1/2 * gamma * S^2 * (sigma_r^2 - sigma_i^2)` each instant. Its total depends on how long the stock stays near the strike, so the mean moves with drift. The quadrature of that integral matches the simulated mean in every row.
- Per path, the gamma-rent sum explains the simulated P&L with a correlation of at least 0.9996.
- Share of paths on which the implied-vol hedger lost money: at most 0.00%. The outcome is uncertain in size, not in sign.



3. Trading costs: what to charge, and when to trade

With a proportional cost of 10 basis points per trade, a seller who charges the Black-Scholes price loses the cost of hedging. Leland (1985) says to price and hedge with `sigma^2 * (1 + sqrt(8/pi) * cost / (sigma * sqrt(dt)))`.

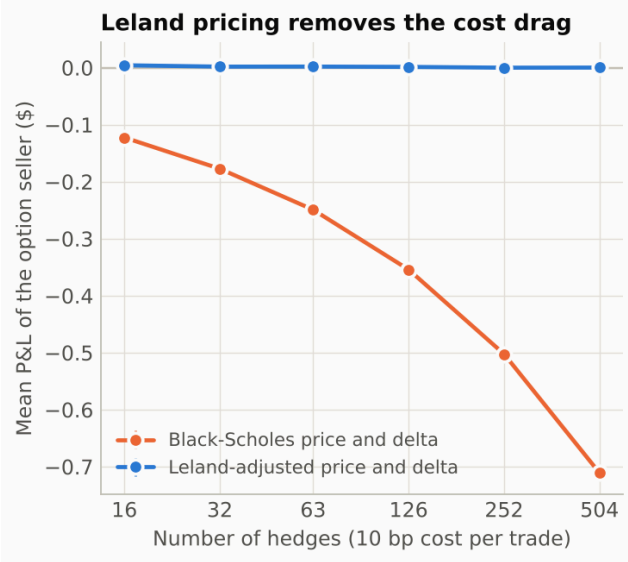
Hedges	Leland number	Adjusted vol	Leland markup	Plain hedge cost	Naive mean	Leland mean	± s.e.
16	0.064	20.63%	0.1253	0.1220	-0.1225	+0.0043	0.0019
32	0.090	20.88%	0.1761	0.1765	-0.1773	+0.0019	0.0014
63	0.127	21.23%	0.2451	0.2496	-0.2492	+0.0020	0.0010
126	0.179	21.72%	0.3425	0.3549	-0.3542	+0.0014	0.0007
252	0.253	22.39%	0.4767	0.5022	-0.5024	+0.0000	0.0005
504	0.358	23.31%	0.6599	0.7113	-0.7110	+0.0002	0.0004

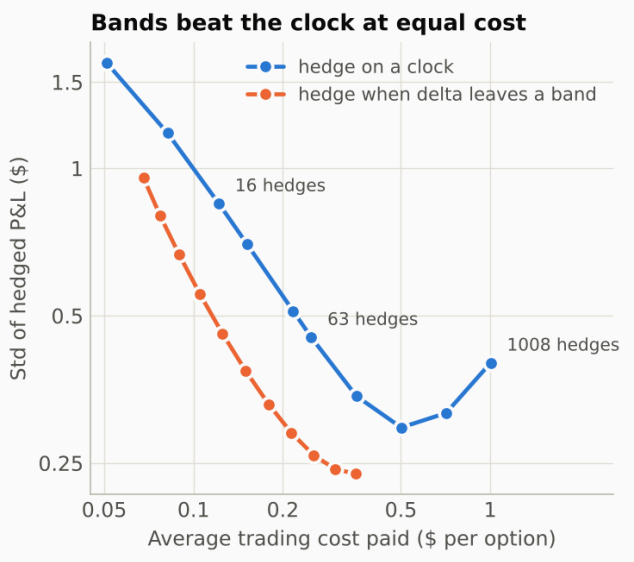
- The adjusted price brings the seller's mean P&L back to within 0.10% of the premium at every frequency tested.
- The markup is what the adjusted hedge costs, not what the unadjusted hedge costs. At 504 hedges the seller who keeps the plain Black-Scholes delta pays 7.8% more than the markup, because hedging at the higher volatility also flattens gamma and so trades less. At 16 hedges the two differ by 2.6%.

Costs make "hedge more often" a trade-off. Two rules were run on the same 40,000 paths: hedge on a clock, or watch continuously and trade only when delta leaves a band (Whalley and Wilmott, 1997), trading back to the band's edge.

Risk aversion	Avg cost	Band std	Clock std at same cost	Risk reduction
0.05	0.0678	0.9562	1.3450	28.9%
0.1	0.0772	0.8004	1.2287	34.9%
0.2	0.0894	0.6665	1.0949	39.1%
0.4	0.1052	0.5525	0.9562	42.2%
0.8	0.1251	0.4598	0.8270	44.4%
1.6	0.1496	0.3865	0.7080	45.4%
3.2	0.1790	0.3299	0.6044	45.4%
6.4	0.2140	0.2879	0.5167	44.3%
12.8	0.2545	0.2592	0.4444	41.7%
25.6	0.3006	0.2429	0.3897	37.7%
51.2	0.3519	0.2377	0.3442	30.9%

At equal average cost the band rule carried 29% to 45% less risk than the clock rule (largest at a cost of $0.18). The clock rule's risk also stops improving past a few hundred hedges, because the cost itself becomes a source of variance. The clock curve is interpolated between tested frequencies, and the band is the small-cost asymptotic formula, not a numerically optimal one.



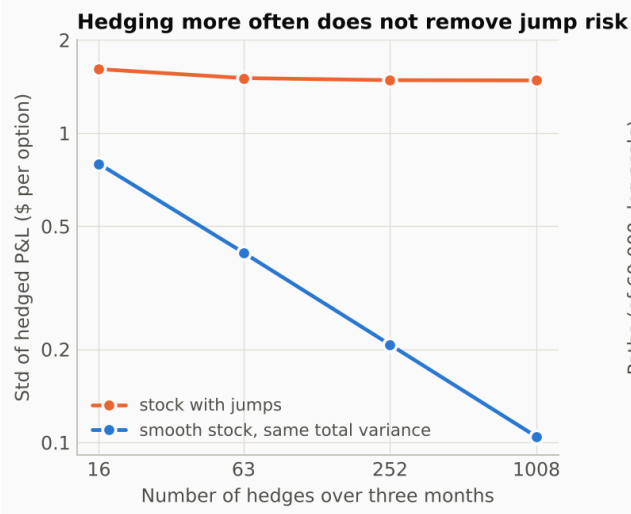


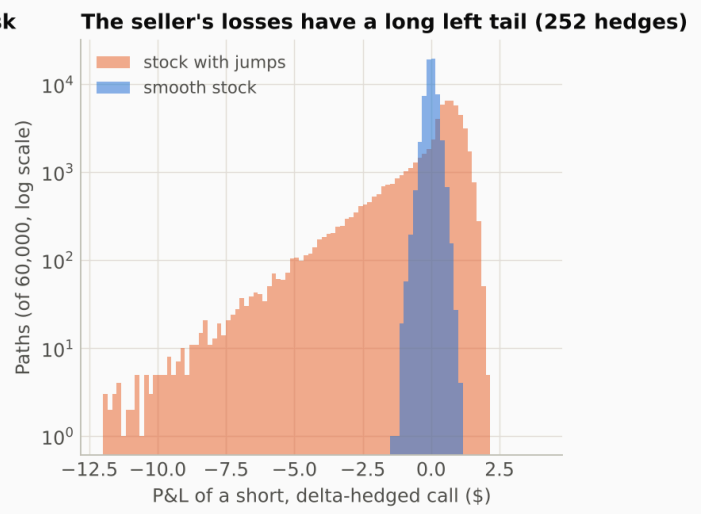
4. Jumps: the risk delta hedging cannot remove

A stock with four jumps a year on average (mean -2%, std 6%) is compared with a smooth stock of the same total variance (18.9% volatility). The option is sold at its fair Merton price, so both sellers break even on average.

Hedges	Smooth std	Jump std	Jump skew	Smooth 1% worst	Jump 1% worst	Jump mean	± s.e.
16	0.796	1.611	-2.09	-2.19	-5.95	-0.0031	0.0036
63	0.411	1.506	-2.42	-1.11	-5.77	+0.0044	0.0034
252	0.207	1.483	-2.53	-0.55	-5.73	+0.0008	0.0033
1008	0.104	1.481	-2.57	-0.28	-5.79	+0.0019	0.0033

Going from 16 to 1008 hedges cut the smooth stock's risk by 87% and the jumping stock's by 8%. At the highest frequency the seller's 1%-worst outcome is 21 times larger with jumps. A gap moves the stock before the hedge can follow; only owning other options removes that risk.





5. From prices to an arbitrage-free smile

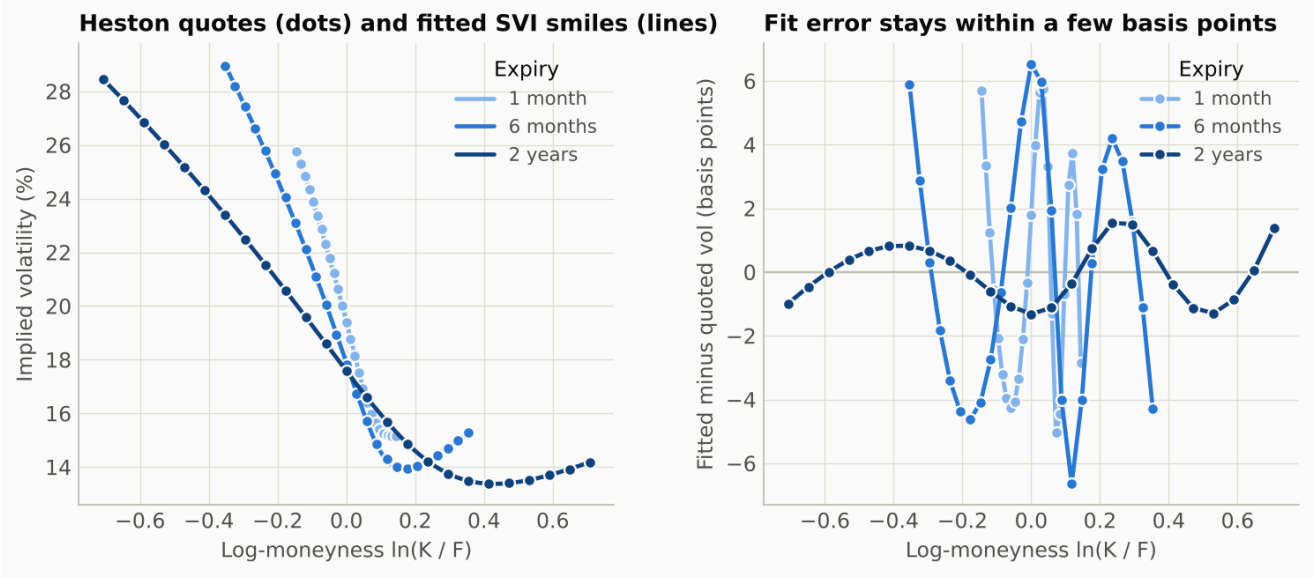
Implied volatility. 1,000,000 random options were priced and inverted at 779,536 options per second on the recorded machine. Accuracy is limited by floating point, not by the solver:

Time value above (× forward)	Options	Solved	Max abs error	Median abs error
0.0001	897,024	897,024	3.0e-13	2.2e-16
1e-06	928,459	928,459	2.6e-11	2.2e-16
1e-08	944,070	944,070	9.4e-10	2.2e-16
1e-10	953,607	953,607	1.2e-07	2.2e-16

Smile fitting. Option chains were generated from a Heston model (kappa 2, theta 0.04, xi 0.6, rho -0.7), five expiries of 25 strikes each, and a raw SVI slice was fitted to each expiry.

Quote noise (vol bp)	Fits	RMSE vs quotes (bp)	RMSE vs true smile (bp)
0	5	2.94	2.94
10	100	9.37	5.44
30	100	26.41	13.58

- Clean quotes are fitted to 2.9 basis points on average (worst expiry 5.8). SVI is not the Heston smile, so the error is not zero.
- With noisy quotes the fit is closer to the true smile than the quotes are: five parameters average out the noise of 25 strikes.
- 0 of 205 fitted slices show butterfly arbitrage inside the quoted strike range, and the worst calendar violation is 0. The test suite confirms the detector does fire on a published arbitrageable smile.



Limitations

- Every result is a simulation under a stated model. Nothing here is a backtest or evidence that a strategy makes money in a real market.
- One option, one maturity, constant rates, no dividends in the hedging studies, no bid-ask spread on the option, and costs proportional to notional.

- The jump and volatility-mismatch studies give the hedger a volatility number it could not know in advance.
- Standard errors describe Monte Carlo noise only. Comparing many table rows to theory was not corrected for multiple comparisons.
- The Leland and Whalley-Wilmott formulas are small-cost approximations; the comparison with them is a check of their accuracy in one configuration.

Reproduce

```
python -m gammascalp research        # about 9 minutes; writes results/
python scripts/build_report.py       # figures and this memo
```

Versions used for the delivered run: Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, Matplotlib 3.11.2. Seeds are fixed in `experiments.py`; other library versions can change the last digits.

References

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